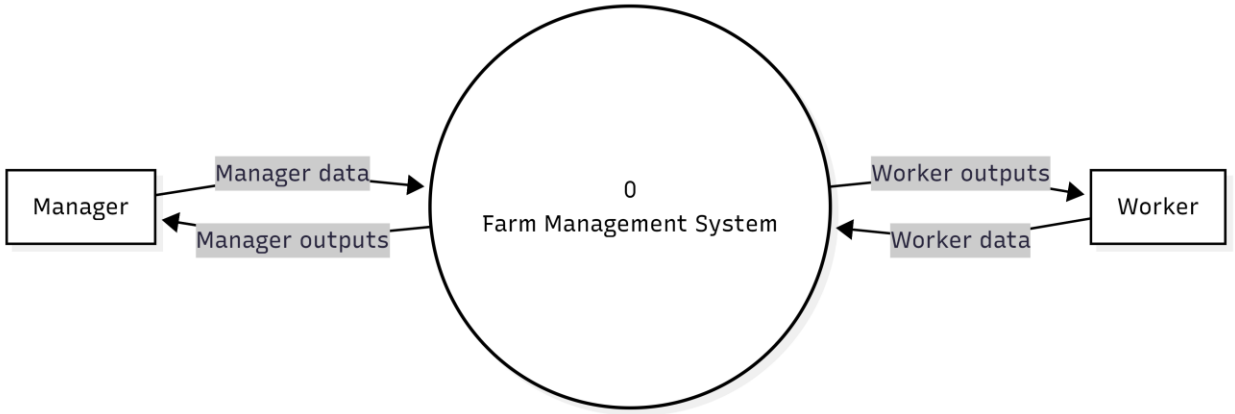
Requirements Document

1. Context Diagram (Level 0 DFD)



Description

The Context Diagram (Level 0 DFD) represents the **Farm Management System** as a single central process that interacts with the external entities that use the system directly.

In this system, the external entities are:

- **Manager**
- **Worker**

The **Manager** sends data to the system such as:

- login information
- farm details
- plot details
- greenhouse details
- plant records
- task creation and assignment information
- task postponement, cancellation, and confirmation data
- harvest records
- expense records
- sales records

- inventory updates
- report requests

The system sends data back to the **Manager** such as:

- farm overview
- plant status information
- task summaries
- worker progress updates
- harvest summaries
- finance summaries
- inventory summaries
- reports and dashboards

The **Worker** sends data to the system such as:

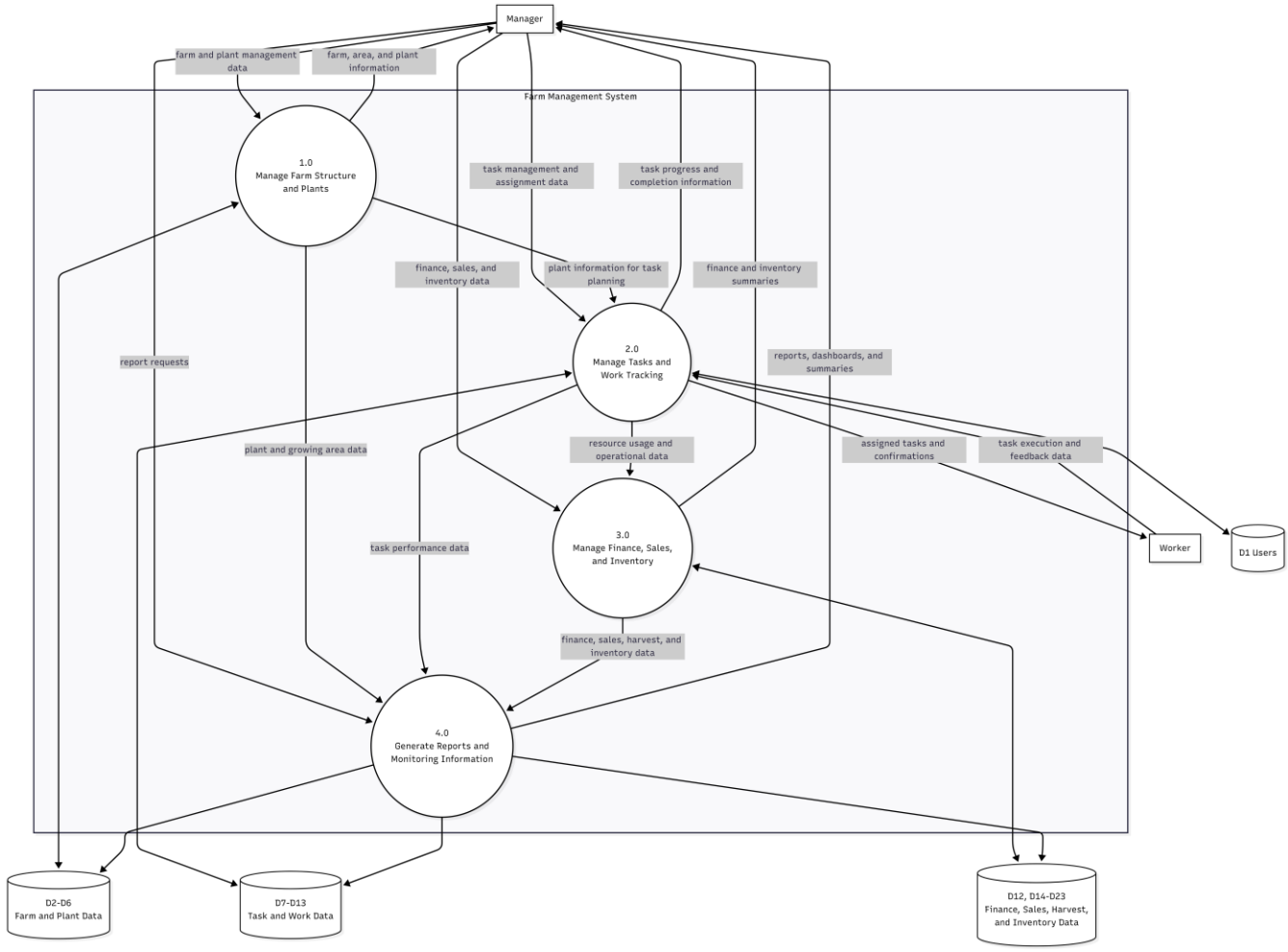
- login information
- task progress updates
- task start updates
- completion updates
- comments
- issue or delay reports
- resource usage records

The system sends data back to the **Worker** such as:

- assigned task list
- task schedules
- task details
- plant-related task information
- task status confirmations

This diagram shows the overall system boundary and the top-level flow of data between the system and its users. It reflects the real application scope, where only managers and workers interact directly with the system.

2. Level 1 DFD



Description

The Level 1 DFD decomposes the main Farm Management System into smaller internal processes. The external entities remain the same:

- **Manager**
- **Worker**

The system is divided into the following main processes:

- **1.0 Manage Farm Structure and Plants**
- **2.0 Manage Tasks and Work Tracking**
- **3.0 Manage Finance, Sales, and Inventory**

- **4.0 Generate Reports and Monitoring Information**

The main data stores used in the system are:

- Users
- Farms
- Plots
- Greenhouses
- Plant Stages
- Plants
- Recurring Task Series
- Tasks
- Task Assignments
- Task Comments
- Task History
- Harvest History
- Resource Usage
- Finance Partners
- Expense Categories
- Recurring Expenses
- Expense Records
- Sales Deals
- Sales Deliveries
- Finance Transactions
- Inventory Categories
- Inventory Items
- Inventory Movements

These data stores are based on the database structure of the system.

2.1 Process 1.0 – Manage Farm Structure and Plants

This process is responsible for managing the farm structure and plant-related information.

It includes:

- creating and updating farms
- creating and updating plots
- creating and updating greenhouses
- creating and updating plant records
- updating plant stages and statuses

The **Manager** sends area and plant data into this process.
The process stores and retrieves data from:

- Farms
- Plots
- Greenhouses
- Plant Stages
- Plants

The output of this process includes:

- updated farm information
- updated plant information
- plant status data used by other processes

2.2 Process 2.0 – Manage Tasks and Work Tracking

This process is responsible for task management and work monitoring.

It includes:

- creating tasks
- assigning workers
- managing recurring task series
- updating task progress
- adding task comments
- reporting issues or delays

- confirming task completion
- recording task history
- recording resource usage

The **Manager** sends:

- task creation data
- assignment data
- postponement or cancellation data
- completion confirmation data

The **Worker** sends:

- task start updates
- progress updates
- completion updates
- comments
- issue reports
- resource usage data

This process uses:

- Recurring Task Series
- Tasks
- Task Assignments
- Task Comments
- Task History
- Resource Usage
- Plants

The output includes:

- assigned tasks for workers
- task status updates for managers
- work progress information

- completion information

2.3 Process 3.0 – Manage Finance, Sales, and Inventory

This process is responsible for financial records, sales records, and stock management.

It includes:

- creating expense records
- managing recurring expenses
- creating sales deals
- recording sales deliveries
- creating finance transactions
- managing inventory items
- recording inventory movements

The **Manager** sends:

- expense data
- recurring expense data
- sales data
- inventory updates

This process uses:

- Harvest History
- Resource Usage
- Finance Partners
- Expense Categories
- Recurring Expenses
- Expense Records
- Sales Deals
- Sales Deliveries
- Finance Transactions

- Inventory Categories
- Inventory Items
- Inventory Movements

The output includes:

- expense summaries
- income summaries
- sales summaries
- stock level information
- inventory movement records

2.4 Process 4.0 – Generate Reports and Monitoring Information

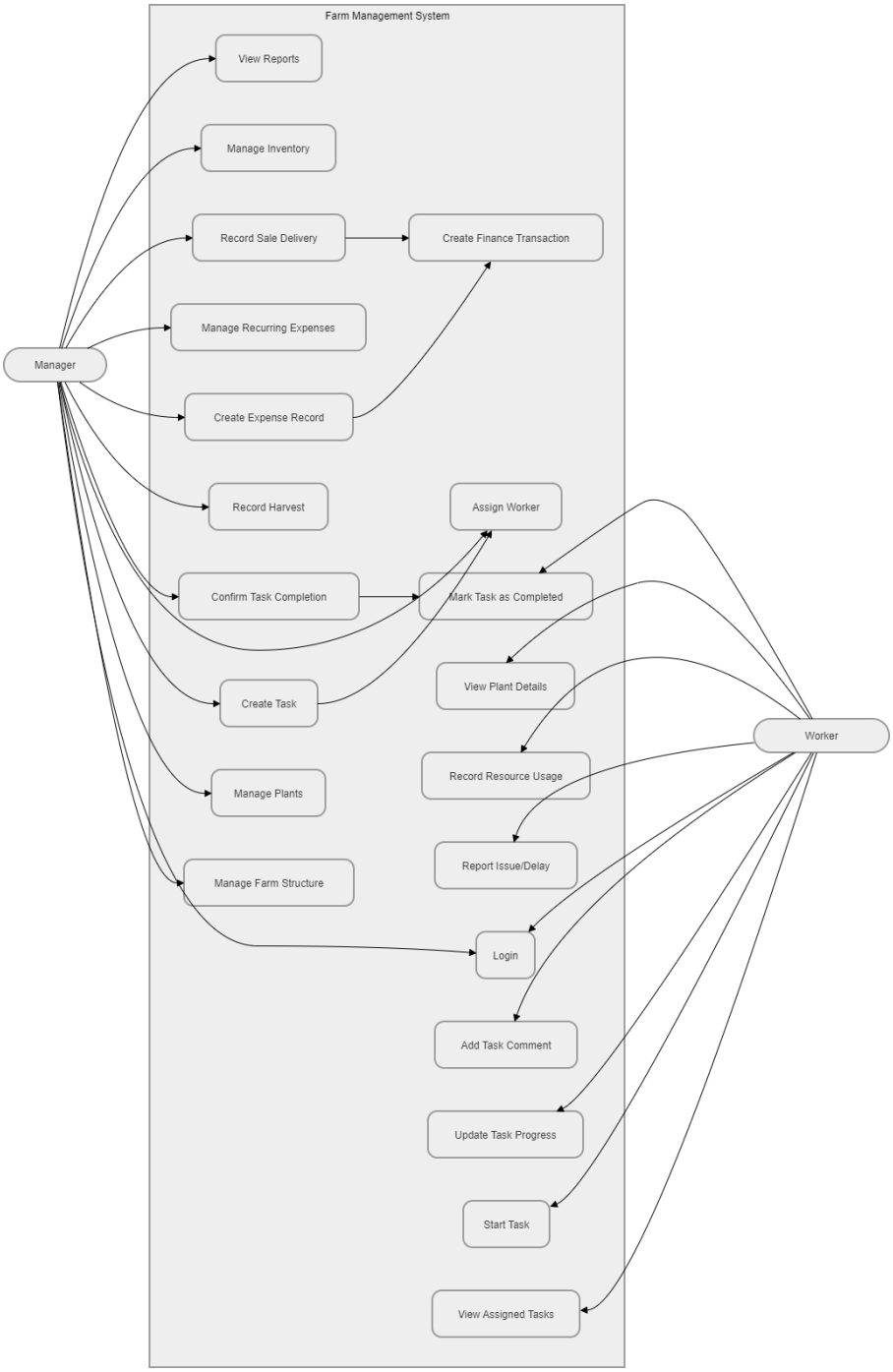
This process is responsible for generating management reports and system summaries.

It collects information from the other internal processes and produces:

- task reports
- harvest reports
- financial reports
- inventory reports
- monthly summaries
- yearly summaries
- dashboards

The **Manager** sends a report request to this process.
The system returns the requested reports and monitoring information.

3. Use Case Diagram



Description

The Use Case Diagram represents the interactions between the users of the system and the main functions of the Farm Management System.

The actors are:

- **Manager**
- **Worker**

The **Manager** is connected to the following use cases:

- Log In
- Manage Farms
- Manage Plots
- Manage Greenhouses
- Manage Plants
- Create Task
- Assign Worker
- Postpone Task
- Cancel Task
- Confirm Task Completion
- Record Harvest
- Create Expense Record
- Manage Recurring Expenses
- Create Sales Deal
- Record Sale Delivery
- Manage Inventory
- View Reports

The **Worker** is connected to the following use cases:

- Log In
- View Assigned Tasks
- Start Task
- Update Task Progress
- Add Task Comment

- Report Issue/Delay
- Record Resource Usage
- Mark Task as Completed
- View Plant Details

Use Case Relationships

The diagram includes the following <<include>> relationships:

- **Create Task** <<include>> **Assign Worker**
- **Create Expense Record** <<include>> **Create Finance Transaction**
- **Record Sale Delivery** <<include>> **Create Finance Transaction**

The diagram also includes the following <<extend>> relationship:

- **Confirm Task Completion** <<extend>> **Mark Task as Completed**

This use case diagram reflects the actual roles and functions supported by the application.

4. Use Case Narratives

4.1 Use Case Narrative 1 – Create and Assign Task

Name:
Create and Assign Task

Summary:
The manager creates a new task for a specific plant and assigns one or more workers so that farm work can be completed on time.

Actors:
Primary Actor: Manager
Secondary Actor: Worker

Preconditions:

- The manager is logged into the system.
- The plant already exists in the system.
- At least one worker account exists and is active.

Main Sequence:

1. The manager opens the Tasks section.
2. The manager selects the option to create a new task.
3. The system displays the task creation form.
4. The manager enters the task title, description, category, priority, scheduled start time, and scheduled end time.
5. The manager selects the related plant.
6. The manager enters required items or additional notes if needed.
7. The manager selects one or more workers to assign to the task.
8. The manager submits the task.
9. The system validates the entered data.
10. The system creates the task with status **Scheduled**.
11. The system creates the task assignment records.
12. The system stores a task history record.
13. The system displays a success message.
14. The assigned workers can now see the task in their task list.

Alternative Sequences:
A1. Invalid schedule

- If the scheduled end time is earlier than the scheduled start time, the system rejects the submission.
- The system highlights the invalid fields.
- The manager corrects the time values and submits again.

A2. Plant not available

- If the selected plant is inactive, removed, or invalid, the system prevents task creation.
- The manager must select another valid plant.

A3. Worker assignment problem

- If a selected worker is inactive or invalid, the system shows an error.
- The manager removes or replaces that worker before continuing.

Postcondition:
A new scheduled task is stored in the system, linked to a plant, assigned to workers, and visible in the workers' task list.

4.2 Use Case Narrative 2 – Update Task Progress and Confirm Completion

Name:
Update Task Progress and Confirm Completion

Summary:
A worker updates the progress of an assigned task and marks it as completed. The manager then reviews and confirms the completion.

Actors:
Primary Actor: Worker
Secondary Actor: Manager

Preconditions:

- The worker is logged into the system.
- The worker is assigned to the task.
- The task exists and is in a valid working status.

Main Sequence:

1. The worker opens the Assigned Tasks page.
2. The worker selects a task.
3. The worker starts the task.
4. The system changes the task status to **In Progress** and records the start time.
5. The worker performs the work.
6. The worker adds notes, comments, or resource usage if needed.
7. The worker selects **Mark as Completed**.
8. The system changes the task status to **Completed Pending Confirmation**.
9. The system records the worker completion time.
10. The manager opens the task review page.
11. The manager reviews the completion details.

12. The manager selects **Confirm Completion**.
13. The system changes the task status to **Completed**.
14. The system stores the manager confirmation time.
15. The system records the change in task history.

Alternative Sequences:
B1. Worker reports an issue or delay

- Instead of completing the task, the worker enters an issue or delay note.
- The system stores the comment in task comments/history.
- The task remains **In Progress** until the issue is resolved.

B2. Manager rejects completion

- The manager decides that the work is incomplete or incorrect.
- The manager enters a note explaining the reason.
- The system changes the task back to **In Progress** or **Postponed**.
- The action is stored in task history.

B3. Task is cancelled before completion

- Before completion, the manager cancels the task.
- The system changes the status to **Cancelled** and stores the reason.
- The worker can no longer complete that task.

Postcondition:
The task is either confirmed as completed, returned for more work, postponed, or cancelled, and all status changes are recorded.

4.3 Use Case Narrative 3 – Create Expense Record

Name:
Create Expense Record

Summary:
The manager records a farm expense so that the system can store the cost, payment status, and related finance information for tracking and reporting.

Actors:
Primary Actor: Manager

Preconditions:

- The manager is logged into the system.
- The farm exists in the system.
- A valid expense category exists.

Main Sequence:

1. The manager opens the Finance section.
2. The manager selects **Create Expense Record**.
3. The system displays the expense entry form.
4. The manager selects the farm and expense category.
5. The manager enters the expense title, description, amount, expense date, and payment status.
6. The manager optionally links the expense to a plant, task, recurring expense, or worker wage record.
7. The manager submits the form.
8. The system validates the entered data.
9. The system stores the expense record.
10. The system creates the related finance transaction as an **Expense** entry.
11. The system updates financial summaries.
12. The system displays a confirmation message.

Alternative Sequences:
C1. Invalid amount

- If the amount is missing, zero, or negative, the system rejects the form.
- The manager corrects the amount and submits again.

C2. Invalid due date

- If the due date is earlier than the expense date, the system rejects the entry.
- The manager corrects the dates and submits again.

C3. Invalid linked record

- If the selected task, plant, or recurring expense does not exist or is invalid, the system prevents the link from being saved.
- The manager removes or replaces the invalid link.

Postcondition:
The expense record is stored in the system, a related finance transaction is created, and the new expense appears in finance summaries and reports.